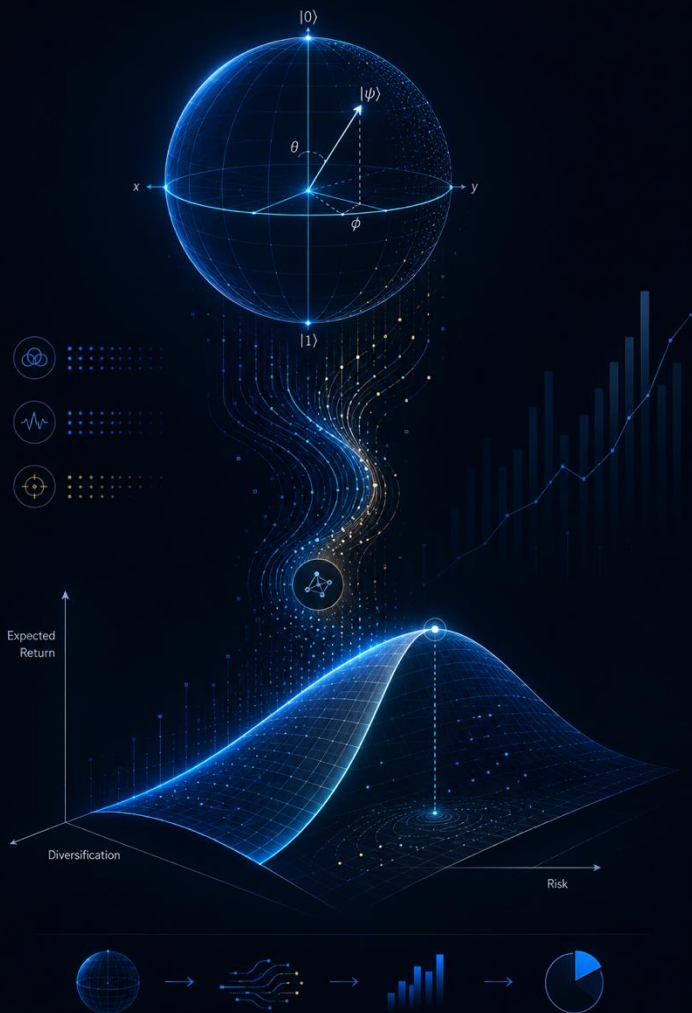




QUANTUM QUANTITATIVE FINANCE

Constrained Portfolio Optimization: A Quantum Interior-Point Approach

A practical walkthrough of classical and quantum continuous portfolio optimization with constraints



Go to:

github.com/Nadav138/QApp/blob/main/workshop/quantum_portfolio_tutorial.ipynb

And open locally or via Google Colab (Recommended)

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The Problem

You have money to spread across many investments. **How much goes into each one?**

The Trade-off

Aim for higher return $\mu^T w$ $\Leftrightarrow$ **Keep risk low** $w^T \Sigma w$

We minimize variance $w^T \Sigma w$ for a required return $\mu^T w = R$.

The feasible region is defined by the constraints:

Use all of it

Invest the whole pot.

$$\sum w_i = 1$$

No short selling

Non-negative amounts only.

$$w_i \geq 0$$

Diversification

Cap any single holding.

$$w_i \leq 0.35$$

Hit a goal

Meet a target return.

$$\mu^T w = R$$

Interior-Point Methods: The Main Idea

This portfolio problem can be written as a convex **Second-Order Cone Program (SOCP)**

A standard form for constrained optimization.

- Constraints define the *feasible region*
- IPM starts inside it and avoids the boundaries using *barrier terms*
- Each iteration solves a **Newton linear system**
- In the notebook:
 - **CVXPY** models the SOCP
 - **CLARABEL** solves it with a classical IPM

LOG-BARRIER

$$-\tau \sum_i \log(w_i)$$

Example: non-negativity barrier

Discourages approaching $w_i = 0$

IPM turns constrained optimization into a sequence of Newton **linear systems**

From Interior-Point to Quantum Interior-Point

CLASSICAL IPM

Start at a feasible interior point



Build the Newton system $K \Delta z = r$



Solve Newton system classically



Update variables · repeat until the duality gap is small

QUANTUM IPM (QIPM)

Same IPM outer loop



Same Newton system $K \Delta z = r$



Solve with HHL after Hermitian dilation



Extract an approximate update

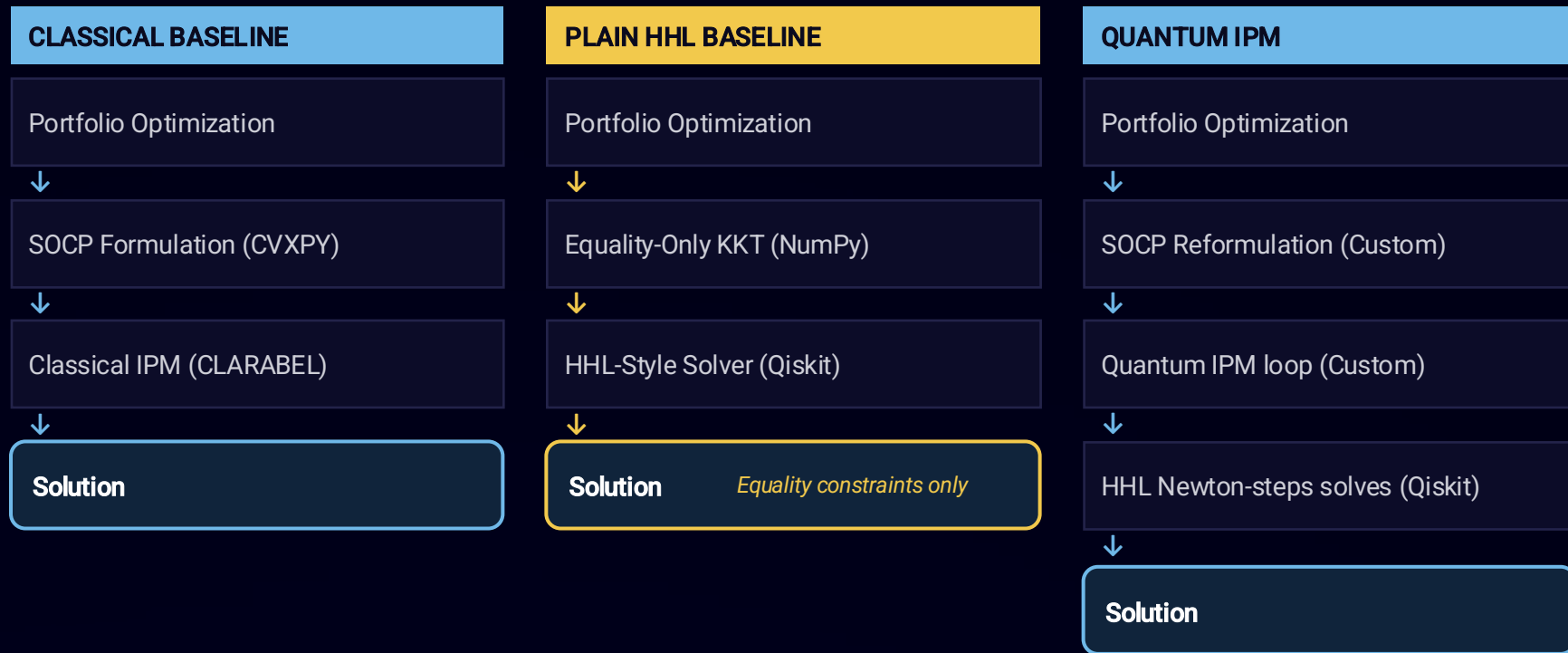


Repeat until the duality gap is small

Swaps the inner engine, not the optimization.

Potential speedups come from faster Newton-system solves under suitable assumptions.

Three Pipelines, One Story



How We Evaluate the Demo

Beyond the paper's theory, we add **practical portfolio metrics**.

1. Constraint Satisfaction

- Budget: $\sum w_i = 1$
- Long-only & cap: $0 \leq w_i \leq w_{max}$
- Return target: $\mu^T w = R$

2. Risk, Return & Sharpe

- Expected return: $\mu^T w$
- Variance: $w^T \Sigma w$
- Sharpe $\approx$ *annualized return / annualized volatility*

3. Out-of-Sample Performance

- Train on 2024, test on unseen 2025 data
- Report total return on the held-out year

4. Diversification

- HHI = $\sum_i w_i^2$
- Range for long-only, fully invested: $1/N \leq HHI \leq 1$
- Lower HHI = less concentration

The Notebook Walkthrough — What We'll Build

1.

Market Data

Download 2024 prices and compute returns and covariance.

2.

Classical Baseline

Solve the constrained portfolio with CVXPY. Show what breaks when inequalities are removed.

3.

Equality-Only HHL

4.

SCOP + Quantum IPM

Reformulate constraints
Use HHL-style solves inside the IPM loop.

5.

Out-of-Sample Test + Analysis

Evaluate the portfolios on unseen 2025 data.

Let's dive into the code!



Go to: github.com/Nadav138/QApp/blob/main/workshop/quantum_portfolio_tutorial.ipynb

THANK YOU

Questions?

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Applications of Quantum Computing
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